Project

Om Badhe

2/28/23

```
library(ggplot2)
library(stringr)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(tidyverse)
```

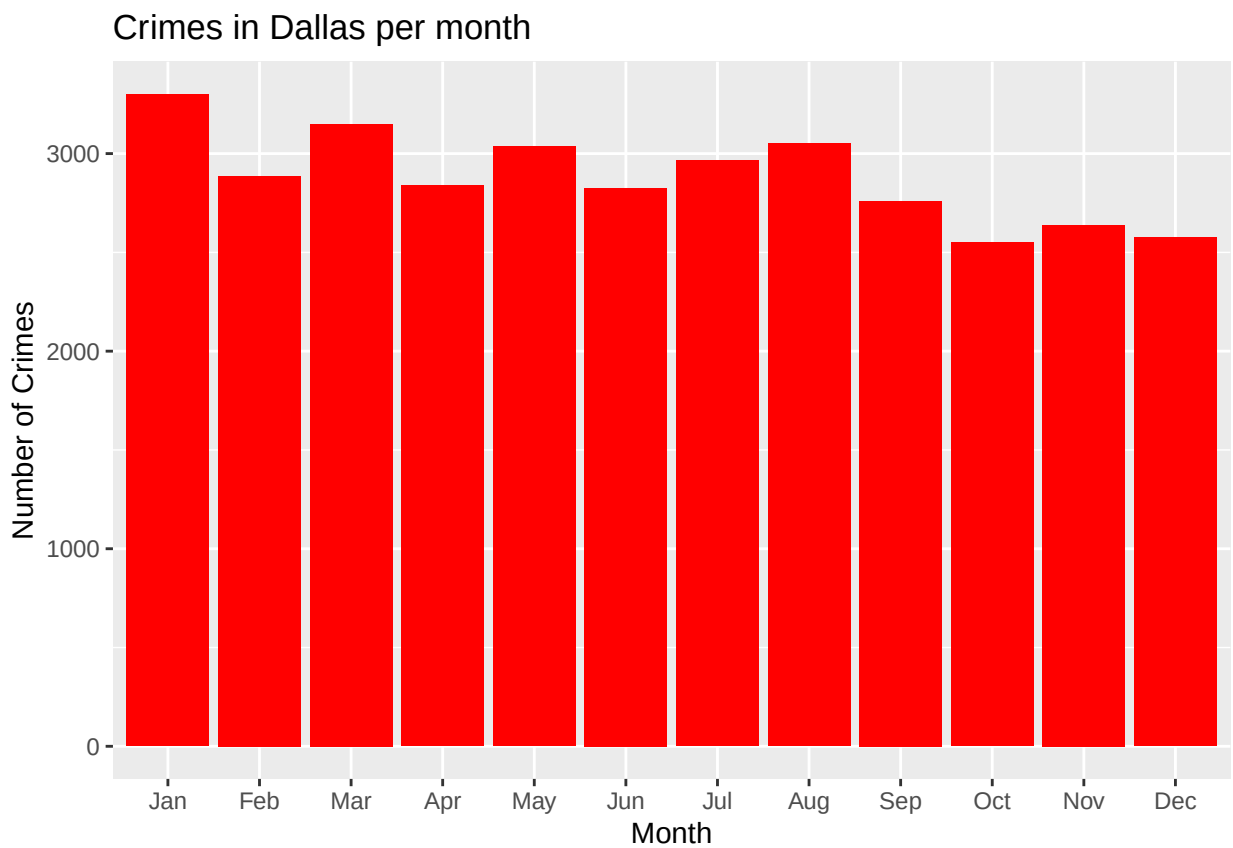
```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2
## --

## v tibble  3.1.8      v purrr   1.0.1
## v tidyr   1.3.0      v dplyr   1.0.10
## v readr   2.1.3      v forcats 0.5.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x lubridate::as.difftime() masks base::as.difftime()
## x lubridate::date()        masks base::date()
## x dplyr::filter()          masks stats::filter()
## x lubridate::intersect()   masks base::intersect()
## x dplyr::lag()             masks stats::lag()
## x lubridate::setdiff()     masks base::setdiff()
## x lubridate::union()       masks base::union()
```

```
library(dplyr)
library(base)
dallas_crimes <- read.csv(file = "crime.csv", header = T)
phoenix_crimes <- read.csv(file = "arrestdetail.csv")
# cleaning the data

# sub-setting the data to get only from 2018-2022
index <- which(dallas_crimes$Arrest.Year >= 2018 & dallas_crimes$Arrest.Year <= 2022)
dallas_crimes <- dallas_crimes[dallas_crimes$Arrest.Year >= 2018 & dallas_crimes$Arrest.Year <= 2022, ]
dallas_crimes <- arrange(dallas_crimes, Arrest.Year, Arrest.Date)
```

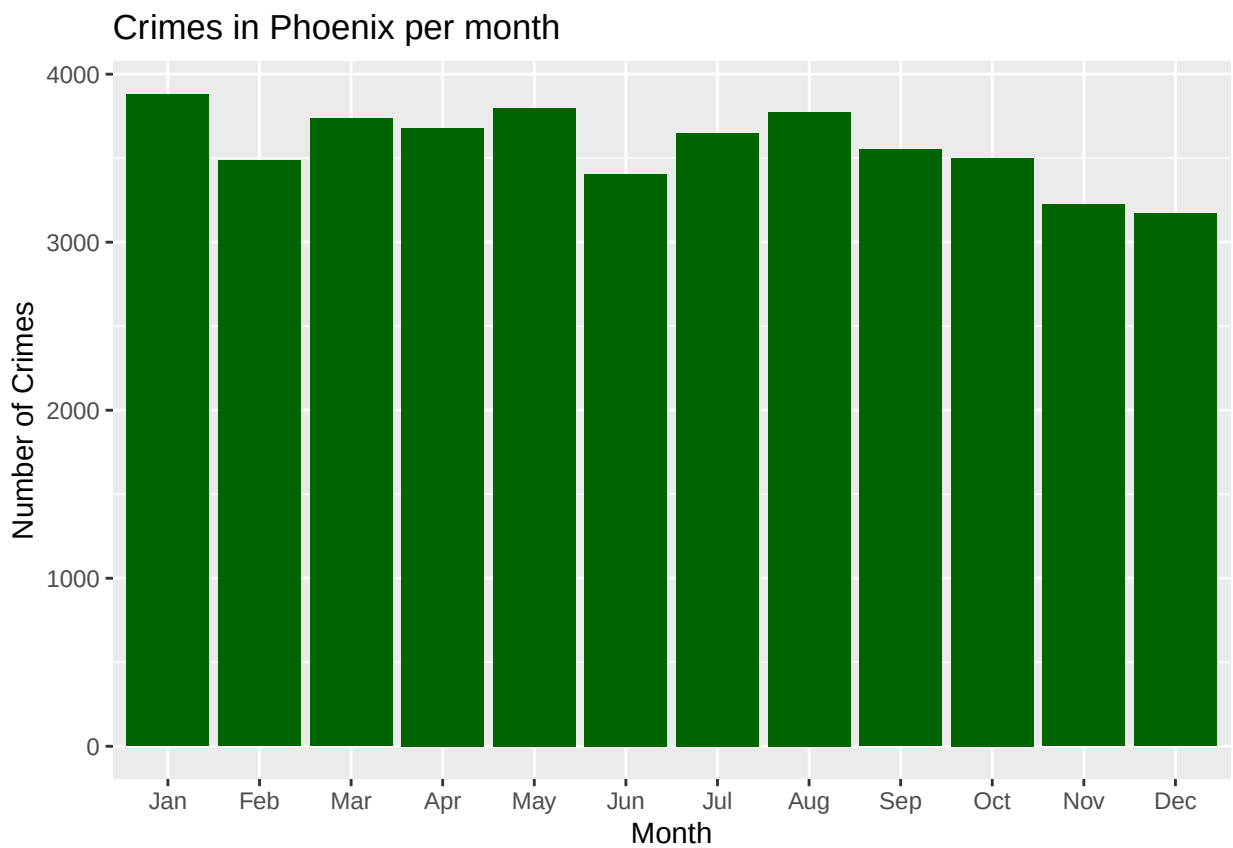
```r
# making the date variable a "as.Date" variable to make it easier for plotting
dallas_crimes$Arrest.Date <- as.Date(dallas_crimes$Arrest.Date, format = "%m/%d/%y", tryFormats = c("%m/

# making the date variable a "as.Date" variable to make it easier for plotting
phoenix_crimes$DATE_OCCUR <- as.Date(phoenix_crimes$DATE_OCCUR, format = "%m/%d/%Y")

# getting rid of variables we won't use
phoenix_crimes <- dplyr::select(phoenix_crimes, c(-QTR, -UNIQUE_NAME_ID, -ARST_OFFICER,
                        -PRECINCT_NUM, -BEAT_NUM, -MAPGRID, -COUNCIL_DISTRICT))
# not counting the data in 2023 to make the comparisons cleaner
phoenix_crimes <- filter(phoenix_crimes, YEAR < 2023)

# omitting any "NA's"
phoenix_crimes <- na.omit(phoenix_crimes)
phoenix_crimes <- phoenix_crimes[seq(1, nrow(phoenix_crimes),4), ]

# renaming the column names
colnames(dallas_crimes) <- c("incidentnum", "year", "date", "time",
                             "address", "zipcode", "latitude", "longitude", "city",
                             "state", "dotw", "location", "weapon", "age",
                             "race", "sex", "drugrelated", "type")
colnames(phoenix_crimes) <- c("incidentnum", "date", "dotw", "month", "year",
                               "type", "sex", "race", "ethnicity", "racegrp", "age",
                               "agegrp", "officersex", "officerrace", "officergrp",
                               "address", "precinct", "beat", "felonycharges", "misdemeanorcharges", "oth

# Cleaning the day of the week and month data for Phoenix
rep_str <- c("1-" = "","2-" = "", "3-" = "", "4-" = "", "5-" = "", "6-" = "", "7-" = "")
phoenix_crimes$dotw <- str_replace_all(phoenix_crimes$dotw, rep_str)

rep_str2 <- c("0-" = "", "1-" = "","2-" = "", "3-" = "", "4-" = "", "5-" = "", "6-" = "", "7-" = "", "8-
phoenix_crimes$month <- str_replace_all(phoenix_crimes$month, rep_str2)
phoenix_crimes$month <- str_replace(phoenix_crimes$month,"0", "")
phoenix_crimes$month <- str_replace(phoenix_crimes$month,"1", "")

# Specifying "Other" in the race category
phoenix_crimes$racegrp <- str_replace(phoenix_crimes$racegrp, "Other", "Asian")


# Cleaning the month data for Dallas
dallas_crimes$month <- month(ymd(dallas_crimes$date))
dallas_crimes$month <- month.abb[dallas_crimes$month]
dallas_crimes<- dallas_crimes[order(dallas_crimes$date), ]
dallas_crimes <- dallas_crimes[-c(1,2), ]
capwords <- function(s, strict = FALSE) {
    cap <- function(s) paste(toupper(substring(s, 1, 1)),
                  {s <- substring(s, 2); if(strict) tolower(s) else s},
                             sep = "", collapse = " " )
    sapply(strsplit(s, split = " "), cap, USE.NAMES = !is.null(names(s)))
}

# Abbreviating the data for month and day of the week
phoenix_crimes$month <- substr(phoenix_crimes$month, 1, 3)
```

```r
phoenix_crimes$month <- capwords(tolower((phoenix_crimes$month)))
phoenix_crimes$month <- as.factor(phoenix_crimes$month)
phoenix_crimes$dotw <- substr(phoenix_crimes$dotw, 1, 3)
phoenix_crimes$dotw <- capwords(tolower(phoenix_crimes$dotw))

# Manipulating the data to show yes or no for drug charges based on the number of charges
indexyes <- which(phoenix_crimes$drugcharges > 0)
indexno <- which(phoenix_crimes$drugcharges == 0)
phoenix_crimes$drugcharges[indexyes] <- "yes"
phoenix_crimes$drugcharges[indexno] <- "no"

# Cleaning the weapon variable in Dallas data to only keep relevant ones and combine
# similar categories
index_firearm <- which(dallas_crimes$weapon == 'Firearm (Type Not Stated)' |
                         dallas_crimes$weapon == 'Other Firearm' |
                         dallas_crimes$weapon == 'Gun' |
                         dallas_crimes$weapon == '33')
index_unarmed <- which(dallas_crimes$weapon == 'Unarmed' |
                         dallas_crimes$weapon == 'None' |
                         dallas_crimes$weapon == 'Hands/Feet')
index_blade <- which(dallas_crimes$weapon == 'Knife - Butcher' |
                       dallas_crimes$weapon == 'Knife - Other' |
                       dallas_crimes$weapon == 'Knife - Pocket' |
                       dallas_crimes$weapon == 'Lethal Cutting Instrument' |
                       dallas_crimes$weapon == 'Stabbing Instrument')
index_misc <- which(dallas_crimes$weapon == 'Other' |
                      dallas_crimes$weapon == 'Missle/Arrow' |
                      dallas_crimes$weapon == 'Poison' |
                      dallas_crimes$weapon == 'Rock' |
                      dallas_crimes$weapon == 'Drugs' |
                      dallas_crimes$weapon == 'THREATS')

dallas_crimes$weapon[index_firearm] <- 'Other Firearm'
dallas_crimes$weapon[index_unarmed] <- 'Unarmed'
dallas_crimes$weapon[index_blade] <- 'Blades'
dallas_crimes$weapon[index_misc] <- 'Miscellaneous'


# Getting rid of any Unknowns for racegrp
index <- which(phoenix_crimes$racegrp == "Unknown")

phoenix_crimes <- (phoenix_crimes[-index,])
```

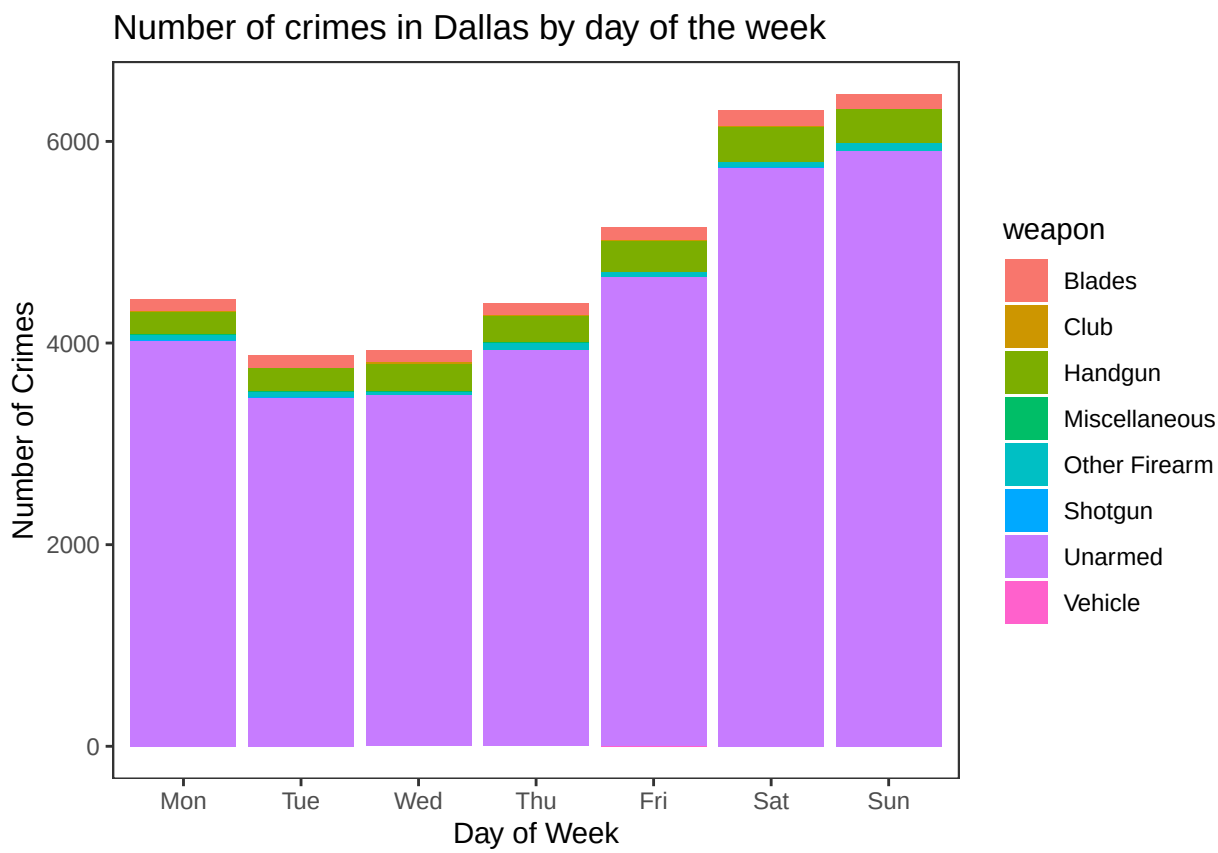
```r
# plotting the Dallas crime per month
ggplot(data = dallas_crimes) +
  geom_bar(aes(x = fct_inorder(month)), fill = "red") +
  ggtitle("Crimes in Dallas per month") +
  labs(x = "Month", y = "Number of Crimes")
```



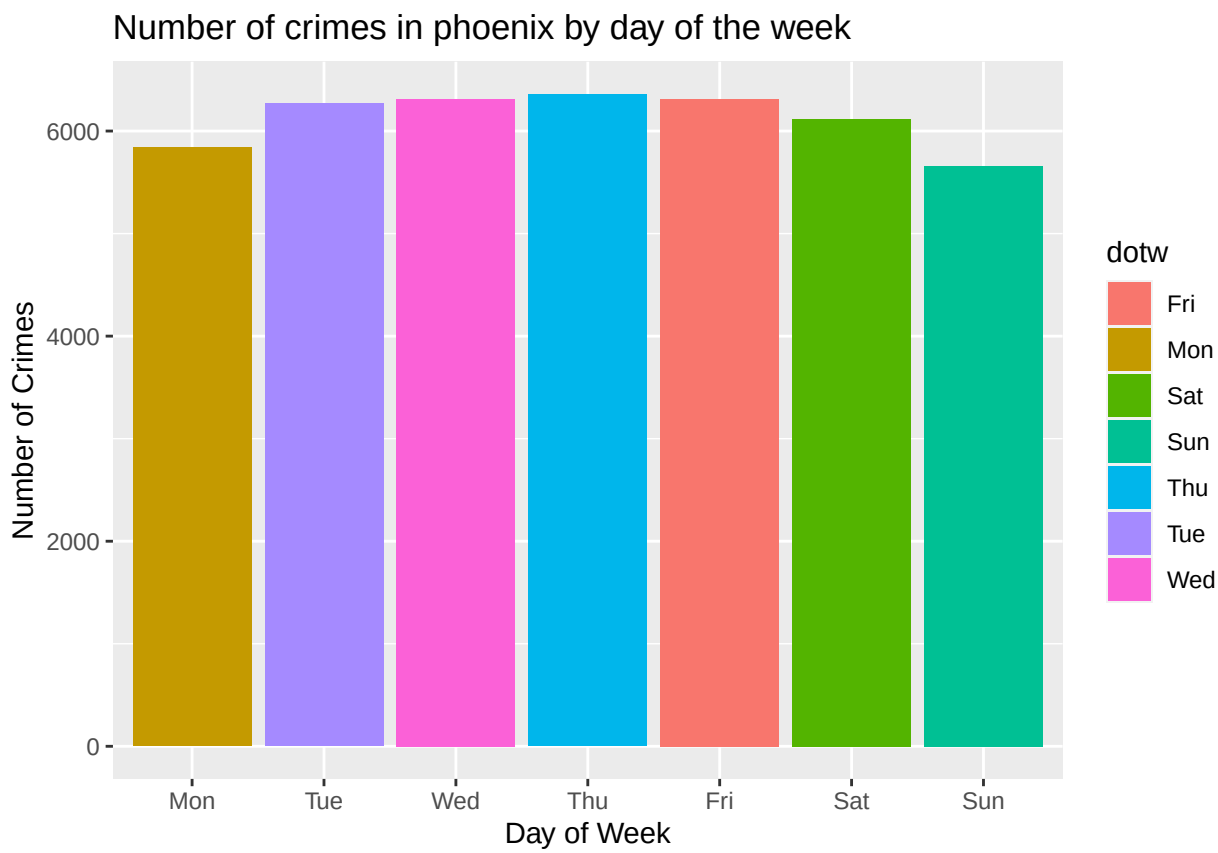

```r
# plotting the Phoenix crime per month
ggplot(data = phoenix_crimes) +
  geom_bar(aes(x = fct_inorder(month)), fill = "darkgreen") +
  ggtitle("Crimes in Phoenix per month") +
  labs(x = "Month", y = "Number of Crimes")
```



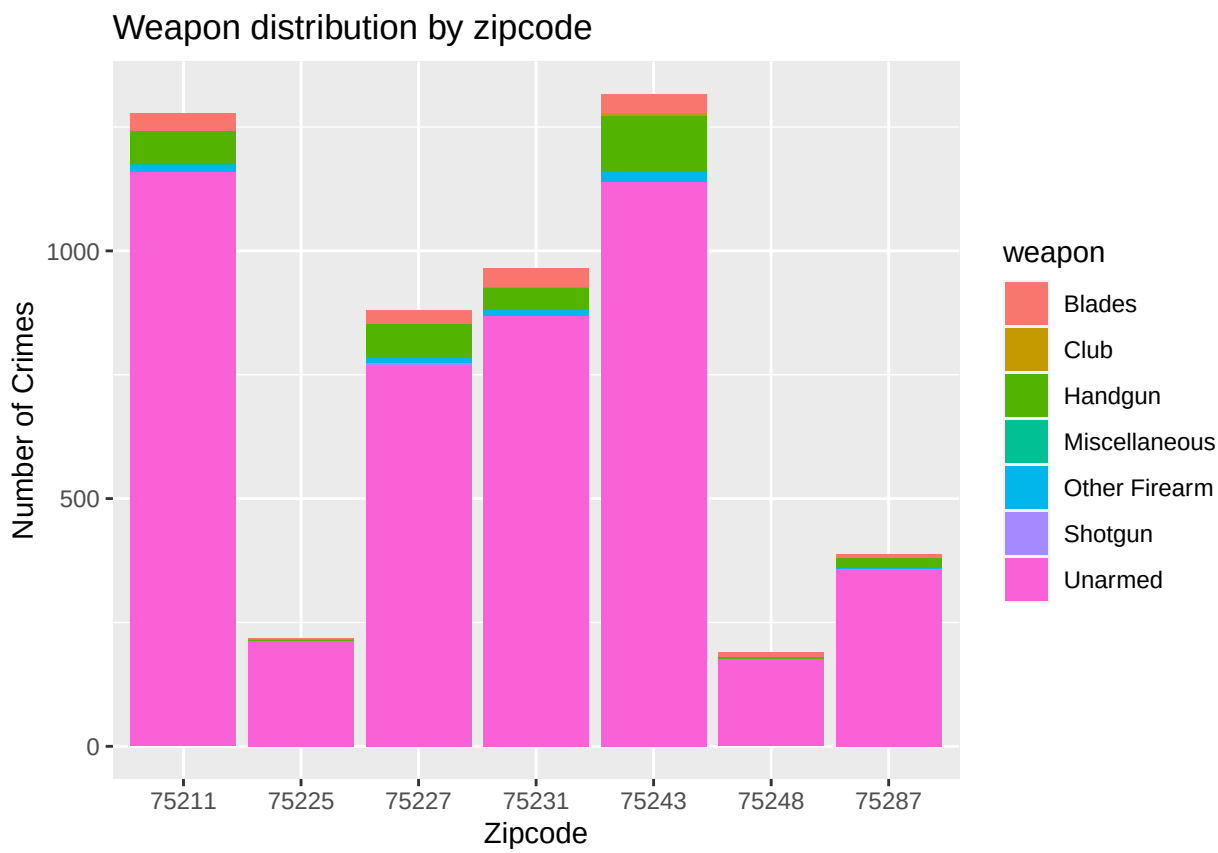

```
# Plotting frequency of crimes during the week for Dallas
ggplot(dallas_crimes) +
  geom_bar(stat = "count", aes(x = fct_inorder(dotw), fill = weapon)) +
  labs(x = "Day of Week", y = "Number of Crimes") +
  ggtitle("Number of crimes in Dallas by day of the week")+
  theme_bw()+
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank())
```




```
# Plotting frequency of crimes during the week for Phoenix

ggplot(phoenix_crimes, aes(x = fct_inorder(dotw))) +
  geom_bar(stat = "count", aes(fill = dotw)) +
  labs(x = "Day of Week", y = "Number of Crimes") +
  ggtitle("Number of crimes in phoenix by day of the week")
```




```
dallas_crimes$zipcode <- as.factor(dallas_crimes$zipcode)
dallas_crimes_zip <- filter(dallas_crimes, zipcode %in% c(75211, 75227, 75243, 75231, 75287, 75225, 752

ggplot(dallas_crimes_zip) +
  geom_bar(stat = 'count', aes(x = (zipcode), fill = weapon)) +
  labs(x = 'Zipcode', y = 'Number of Crimes') +
  ggtitle('Weapon distribution by zipcode')
```

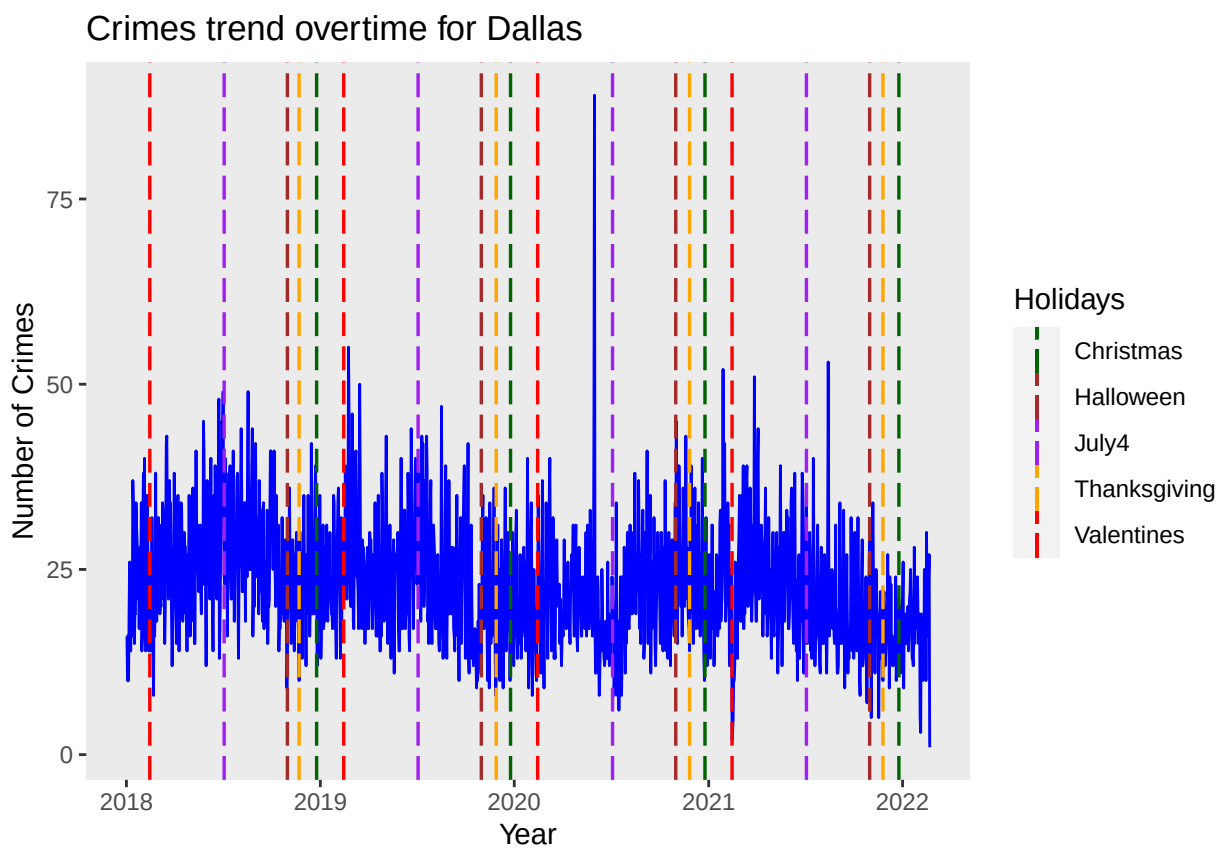


General crime trend... Dallas crime rate may be more sporadic, Phoenix may be more but less spikes. maybe we can add some line to indicate when christmas is or other major holidays

```
######
# Crime in Dallas
ggplot(data = dallas_crimes) +
  geom_line(stat = "count", aes(x = date), color = "blue") +
  ggtitle("Crimes trend overtime for Dallas") +
  labs(x = "Year", y = "Number of Crimes") +
  geom_vline(aes(xintercept = as.Date("2021-12-25"), color = "Christmas"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2020-12-25"), color = "Christmas"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2019-12-25"), color = "Christmas"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2018-12-25"), color = "Christmas"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2021-11-25"), color = "Thanksgiving"), linetype = "longdash", si
  geom_vline(aes(xintercept = as.Date("2020-11-26"), color = "Thanksgiving"), linetype = "longdash", si
  geom_vline(aes(xintercept = as.Date("2019-11-28"), color = "Thanksgiving"), linetype = "longdash", si
  geom_vline(aes(xintercept = as.Date("2018-11-22"), color = "Thanksgiving"), linetype = "longdash", si
  geom_vline(aes(xintercept = as.Date("2021-2-14"), color = "Valentines"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2020-2-14"), color = "Valentines"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2019-2-14"), color = "Valentines"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2018-2-14"), color = "Valentines"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2021-10-31"), color = "Halloween"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2020-10-31"), color = "Halloween"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2019-10-31"), color = "Halloween"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2018-10-31"), color = "Halloween"), linetype = "longdash", size =
  geom_vline(aes(xintercept = as.Date("2021-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
  geom_vline(aes(xintercept = as.Date("2020-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
```

```
  geom_vline(aes(xintercept = as.Date("2019-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
  geom_vline(aes(xintercept = as.Date("2018-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
  scale_color_manual(name = "Holidays", values = c(Christmas = "darkgreen", Thanksgiving = "orange", Val
```

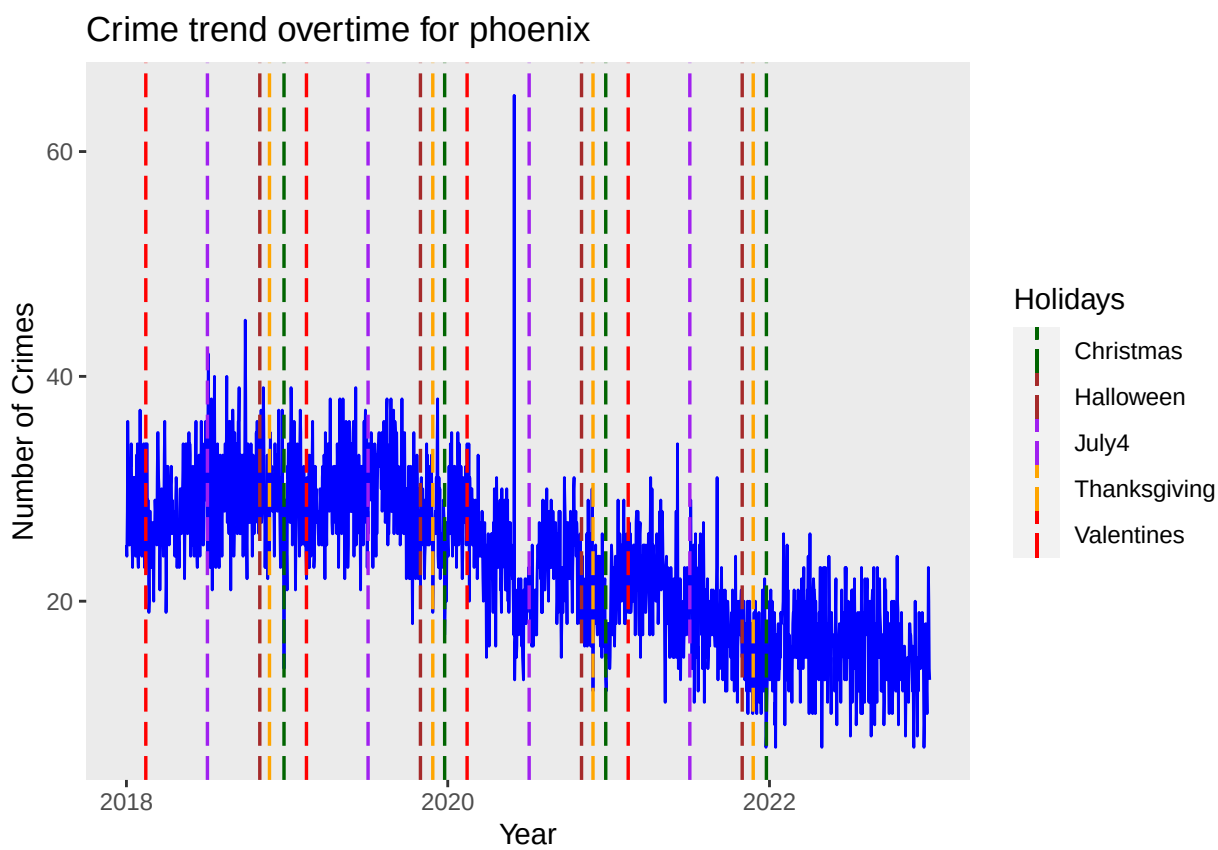
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
```




```
# Comparing crime count on holidays(Dallas)

# Christmas
xmas_18_dal <- length(which(dallas_crimes$date == '2018-12-25'))
xmas_19_dal <- length(which(dallas_crimes$date == '2019-12-25'))
xmas_20_dal <- length(which(dallas_crimes$date == '2020-12-25'))
xmas_21_dal <- length(which(dallas_crimes$date == '2019-12-25'))

# Average crime in Christmas week
xmas_week_18_dal <- floor(length(which(dallas_crimes$date >= "2018-12-22" & dallas_crimes$date <= "2018
xmas_week_19_dal <- floor(length(which(dallas_crimes$date >= "2019-12-22" & dallas_crimes$date <= "2019
xmas_week_20_dal <- floor(length(which(dallas_crimes$date >= "2020-12-22" & dallas_crimes$date <= "2020
xmas_week_21_dal <- floor(length(which(dallas_crimes$date >= "2021-12-22" & dallas_crimes$date <= "2021

# Halloween
hallow_18_dal <- length(which(dallas_crimes$date == '2018-10-31'))
hallow_19_dal <- length(which(dallas_crimes$date == '2019-10-31'))
hallow_20_dal <- length(which(dallas_crimes$date == '2020-10-31'))
```

```
hallow_21_dal <- length(which(dallas_crimes$date == '2021-10-31'))

# Average crime in Halloween Week
hallow_week_18_dal <- floor(length(which(dallas_crimes$date >= "2018-10-28" & dallas_crimes$date <= "201
hallow_week_19_dal <- floor(length(which(dallas_crimes$date >= "2019-10-28" & dallas_crimes$date <= "201
hallow_week_20_dal <- floor(length(which(dallas_crimes$date >= "2020-10-28" & dallas_crimes$date <= "202
hallow_week_21_dal <- floor(length(which(dallas_crimes$date >= "2021-10-28" & dallas_crimes$date <= "202

# July 4th
july4_18_dal <- length(which(dallas_crimes$date == '2018-07-31'))
july4_19_dal <- length(which(dallas_crimes$date == '2019-07-31'))
july4_20_dal <- length(which(dallas_crimes$date == '2020-07-31'))
july4_21_dal <- length(which(dallas_crimes$date == '2021-07-31'))

# Average crime in July 4 week
july4_week_18_dal <- floor(length(which(dallas_crimes$date >= "2018-07-01" & dallas_crimes$date <= "2018
july4_week_19_dal <- floor(length(which(dallas_crimes$date >= "2019-07-01" & dallas_crimes$date <= "2019
july4_week_20_dal <- floor(length(which(dallas_crimes$date >= "2020-07-01" & dallas_crimes$date <= "2020
july4_week_21_dal <- floor(length(which(dallas_crimes$date >= "2018-07-01" & dallas_crimes$date <= "2018

# Thanksgiving
tks_18_dal <- length(which(dallas_crimes$date == '2018-11-22'))
tks_19_dal <- length(which(dallas_crimes$date == '2019-11-28'))
tks_20_dal <- length(which(dallas_crimes$date == '2020-11-26'))
tks_21_dal <- length(which(dallas_crimes$date == '2018-11-25'))

# Average crime in Thanksgiving week
tks_week_18_dal <- floor(length(which(dallas_crimes$date >= "2018-11-19" & dallas_crimes$date <= "2018-
tks_week_19_dal <- floor(length(which(dallas_crimes$date >= "2019-11-25" & dallas_crimes$date <= "2019-
tks_week_20_dal <- floor(length(which(dallas_crimes$date >= "2020-11-23" & dallas_crimes$date <= "2020-
tks_week_21_dal <- floor(length(which(dallas_crimes$date >= "2021-11-22" & dallas_crimes$date <= "2021-

# Valentines
val_18_dal <- length(which(dallas_crimes$date == '2018-02-14'))
val_19_dal <- length(which(dallas_crimes$date == '2019-02-14'))
val_20_dal <- length(which(dallas_crimes$date == '2020-02-14'))
val_21_dal <- length(which(dallas_crimes$date == '2021-02-14'))

# Average crime in Valentines Week
val_week_18_dal <- floor(length(which(dallas_crimes$date >= "2018-02-11" & dallas_crimes$date <= "2018-0
val_week_19_dal <- floor(length(which(dallas_crimes$date >= "2019-02-11" & dallas_crimes$date <= "2019-0
val_week_20_dal <- floor(length(which(dallas_crimes$date >= "2020-02-11" & dallas_crimes$date <= "2020-0
val_week_21_dal <- floor(length(which(dallas_crimes$date >= "2021-02-11" & dallas_crimes$date <= "2021-0

holidays_dallas <- data.frame(holiday = c(xmas_18_dal, xmas_19_dal, xmas_20_dal, xmas_21_dal, hallow_18_
                              weekday = c(xmas_week_18_dal, xmas_week_19_dal, xmas_week_20_dal, xmas_wee
                              name = c(rep('Christmas', 4), rep('Halloween', 4), rep('July4', 4), rep('T
                              year = rep(c('2018', '2019', '2020', '2021'), 5))
affected = c(holidays_dallas$holiday > holidays_dallas$weekday)
holidays_dallas <- cbind(holidays_dallas, affected)
######
# Crime in Phoenix
ggplot(data = phoenix_crimes) +
```

```
geom_line(stat = "count", aes(x = date), color = "blue") +
ggtitle("Crime trend overtime for phoenix") +
labs(x = "Year", y = "Number of Crimes") +
geom_vline(aes(xintercept = as.Date("2021-12-25"), color = "Christmas"), linetype = "longdash", lwd =
geom_vline(aes(xintercept = as.Date("2020-12-25"), color = "Christmas"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2019-12-25"), color = "Christmas"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2018-12-25"), color = "Christmas"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2021-11-25"), color = "Thanksgiving"), linetype = "longdash", si
geom_vline(aes(xintercept = as.Date("2020-11-26"), color = "Thanksgiving"), linetype = "longdash", si
geom_vline(aes(xintercept = as.Date("2019-11-28"), color = "Thanksgiving"), linetype = "longdash", si
geom_vline(aes(xintercept = as.Date("2018-11-22"), color = "Thanksgiving"), linetype = "longdash", si
geom_vline(aes(xintercept = as.Date("2021-2-14"), color = "Valentines"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2020-2-14"), color = "Valentines"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2019-2-14"), color = "Valentines"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2018-2-14"), color = "Valentines"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2021-10-31"), color = "Halloween"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2020-10-31"), color = "Halloween"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2019-10-31"), color = "Halloween"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2018-10-31"), color = "Halloween"), linetype = "longdash", size =
geom_vline(aes(xintercept = as.Date("2021-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
geom_vline(aes(xintercept = as.Date("2020-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
geom_vline(aes(xintercept = as.Date("2019-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
geom_vline(aes(xintercept = as.Date("2018-7-4"), color = "July4"), linetype = "longdash", size = 0.6)
scale_color_manual(name = "Holidays", values = c(Christmas = "darkgreen", Thanksgiving = "orange", Va
```



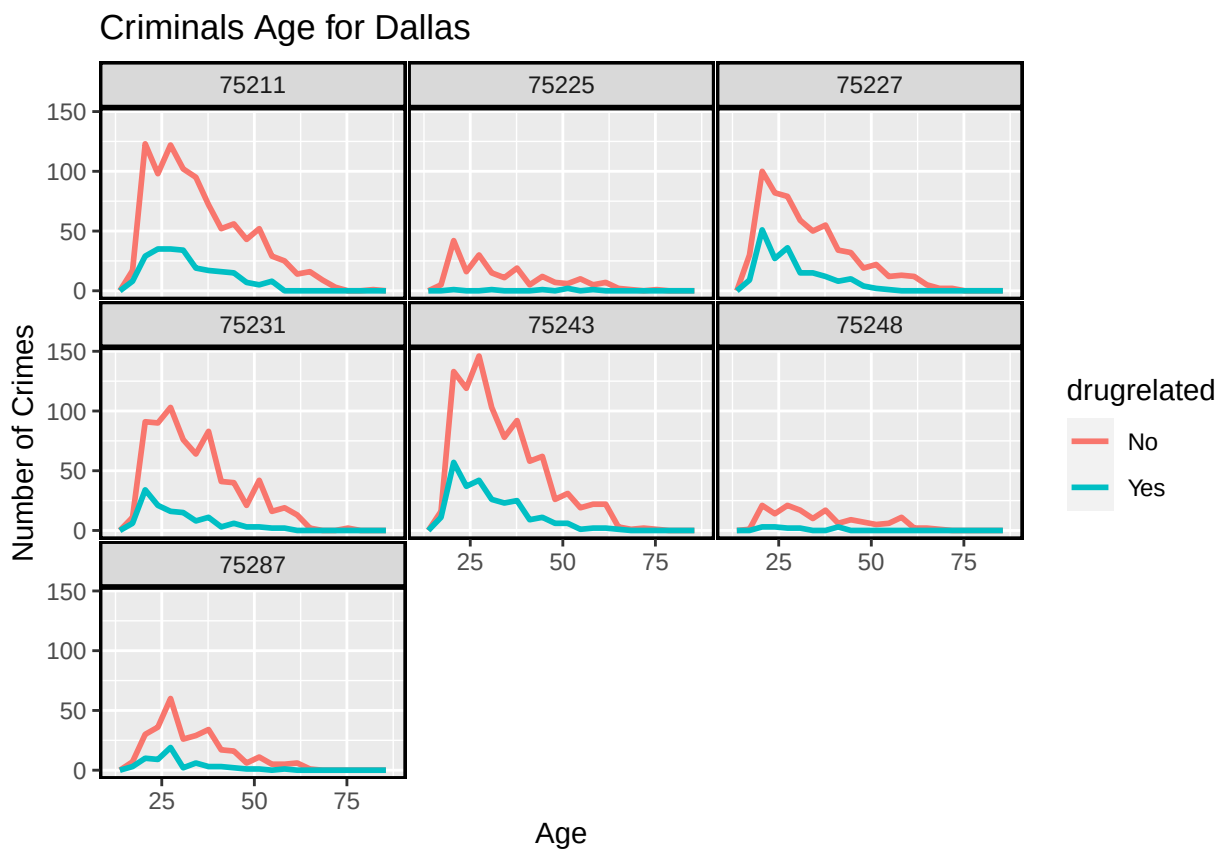
```r
xmas_18_pho <- length(which(phoenix_crimes$date == '2018-12-25'))
xmas_19_pho <- length(which(phoenix_crimes$date == '2019-12-25'))
xmas_20_pho <- length(which(phoenix_crimes$date == '2020-12-25'))
xmas_21_pho <- length(which(phoenix_crimes$date == '2019-12-25'))

# Average crime in Christmas week
xmas_week_18_pho <- floor(length(which(phoenix_crimes$date >= "2018-12-22" & phoenix_crimes$date <= "201
xmas_week_19_pho <- floor(length(which(phoenix_crimes$date >= "2019-12-22" & phoenix_crimes$date <= "201
xmas_week_20_pho <- floor(length(which(phoenix_crimes$date >= "2020-12-22" & phoenix_crimes$date <= "202
xmas_week_21_pho <- floor(length(which(phoenix_crimes$date >= "2021-12-22" & phoenix_crimes$date <= "202

# Halloween
hallow_18_pho <- length(which(phoenix_crimes$date == '2018-10-31'))
hallow_19_pho <- length(which(phoenix_crimes$date == '2019-10-31'))
hallow_20_pho <- length(which(phoenix_crimes$date == '2020-10-31'))
hallow_21_pho <- length(which(phoenix_crimes$date == '2021-10-31'))

# Average crime in Halloween Week
hallow_week_18_pho <- floor(length(which(phoenix_crimes$date >= "2018-10-28" & phoenix_crimes$date <= "2
hallow_week_19_pho <- floor(length(which(phoenix_crimes$date >= "2019-10-28" & phoenix_crimes$date <= "2
hallow_week_20_pho <- floor(length(which(phoenix_crimes$date >= "2020-10-28" & phoenix_crimes$date <= "2
hallow_week_21_pho <- floor(length(which(phoenix_crimes$date >= "2021-10-28" & phoenix_crimes$date <= "2

# July 4th
july4_18_pho <- length(which(phoenix_crimes$date == '2018-07-31'))
july4_19_pho <- length(which(phoenix_crimes$date == '2019-07-31'))
july4_20_pho <- length(which(phoenix_crimes$date == '2020-07-31'))
july4_21_pho <- length(which(phoenix_crimes$date == '2021-07-31'))

# Average crime in July 4 week
july4_week_18_pho <- floor(length(which(phoenix_crimes$date >= "2018-07-01" & phoenix_crimes$date <= "20
july4_week_19_pho <- floor(length(which(phoenix_crimes$date >= "2019-07-01" & phoenix_crimes$date <= "20
july4_week_20_pho <- floor(length(which(phoenix_crimes$date >= "2020-07-01" & phoenix_crimes$date <= "20
july4_week_21_pho <- floor(length(which(phoenix_crimes$date >= "2018-07-01" & phoenix_crimes$date <= "20

# Thanksgiving
tks_18_pho <- length(which(phoenix_crimes$date == '2018-11-22'))
tks_19_pho <- length(which(phoenix_crimes$date == '2019-11-28'))
tks_20_pho <- length(which(phoenix_crimes$date == '2020-11-26'))
tks_21_pho <- length(which(phoenix_crimes$date == '2018-11-25'))

# Average crime in Thanksgiving week
tks_week_18_pho <- floor(length(which(phoenix_crimes$date >= "2018-11-19" & phoenix_crimes$date <= "2018
tks_week_19_pho <- floor(length(which(phoenix_crimes$date >= "2019-11-25" & phoenix_crimes$date <= "2019
tks_week_20_pho <- floor(length(which(phoenix_crimes$date >= "2020-11-23" & phoenix_crimes$date <= "2020
tks_week_21_pho <- floor(length(which(phoenix_crimes$date >= "2021-11-22" & phoenix_crimes$date <= "2021

# Valentines
val_18_pho <- length(which(phoenix_crimes$date == '2018-02-14'))
val_19_pho <- length(which(phoenix_crimes$date == '2019-02-14'))
val_20_pho <- length(which(phoenix_crimes$date == '2020-02-14'))
val_21_pho <- length(which(phoenix_crimes$date == '2021-02-14'))
```

```r
# Average crime in Valentines Week
val_week_18_pho <- floor(length(which(phoenix_crimes$date >= "2018-02-11" & phoenix_crimes$date <= "2018
val_week_19_pho <- floor(length(which(phoenix_crimes$date >= "2019-02-11" & phoenix_crimes$date <= "2019
val_week_20_pho <- floor(length(which(phoenix_crimes$date >= "2020-02-11" & phoenix_crimes$date <= "2020
val_week_21_pho <- floor(length(which(phoenix_crimes$date >= '2021-02-11' & phoenix_crimes$date <= '2021

holidays_phoenix <- data.frame(holiday = c(xmas_18_pho, xmas_19_pho, xmas_20_pho, xmas_21_pho, hallow_18
                               weekday = c(xmas_week_18_pho, xmas_week_19_pho, xmas_week_20_pho, xmas_we
                               name = c(rep('Christmas', 4), rep('Halloween', 4), rep('July4', 4), rep(
                               year = rep(c('2018', '2019', '2020', '2021'), 5))
affected_pho = c(holidays_phoenix$holiday > holidays_phoenix$weekday)
holidays_phoenix <- cbind(holidays_phoenix, affected_pho)
```

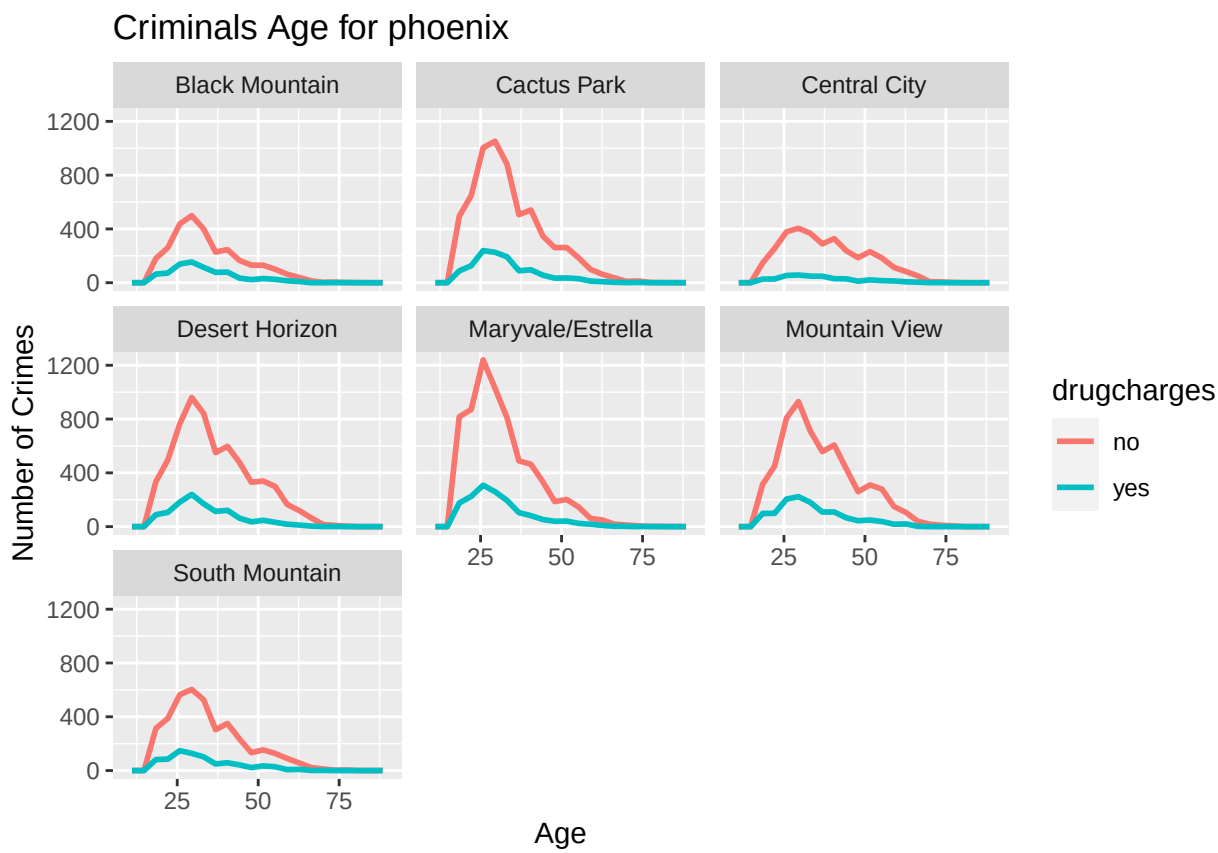
Drug related crimes/ ages committing the crimes

```r
# Getting rid of unknowns and then using some of the relevant zipcodes for Dallas
dallas_crimes <- filter(dallas_crimes, drugrelated != "Uknown")
dallas_crimes_zip <- filter(dallas_crimes, zipcode %in% c(75211, 75227, 75243, 75231, 75287, 75225, 7524
# Plotting the frequency of crime in a zipcode and seeing if they are drug related or not
ggplot(data = dallas_crimes_zip) +
  geom_freqpoly(aes(x = age, color = drugrelated), size = 1, bins = 20) +
  ggtitle("Criminals Age for Dallas") +
  labs(x = "Age", y = "Number of Crimes") +
  facet_wrap(~zipcode) +
    theme(panel.margin=unit(.05, "lines"),
        panel.border = element_rect(color = "black", fill = NA, size = 1),
        strip.background = element_rect(color = "black", size = 1))
```

```
## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.

## Warning: The `panel.margin` argument of `theme()` is deprecated as of ggplot2 2.2.0.
## i Please use the `panel.spacing` argument instead.
```



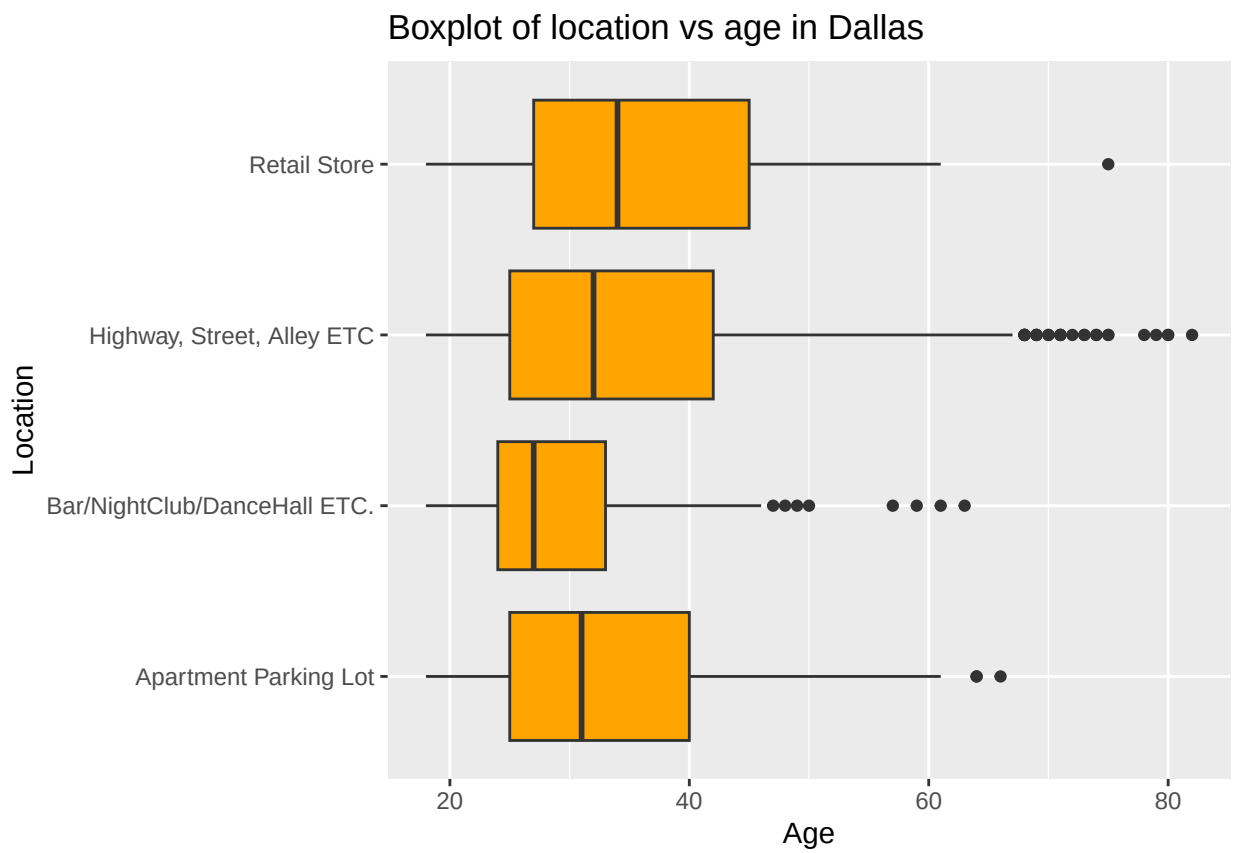

```
# Plotting the frequency of crime in a district and seeing if they are drug related or not

ggplot(data = phoenix_crimes) +
  geom_freqpoly(aes(x = age, color = drugcharges), size = 1, bins = 20) +
  ggtitle("Criminals Age for phoenix") +
  labs(x = "Age", y = "Number of Crimes")+
  facet_wrap(~precinct)
```



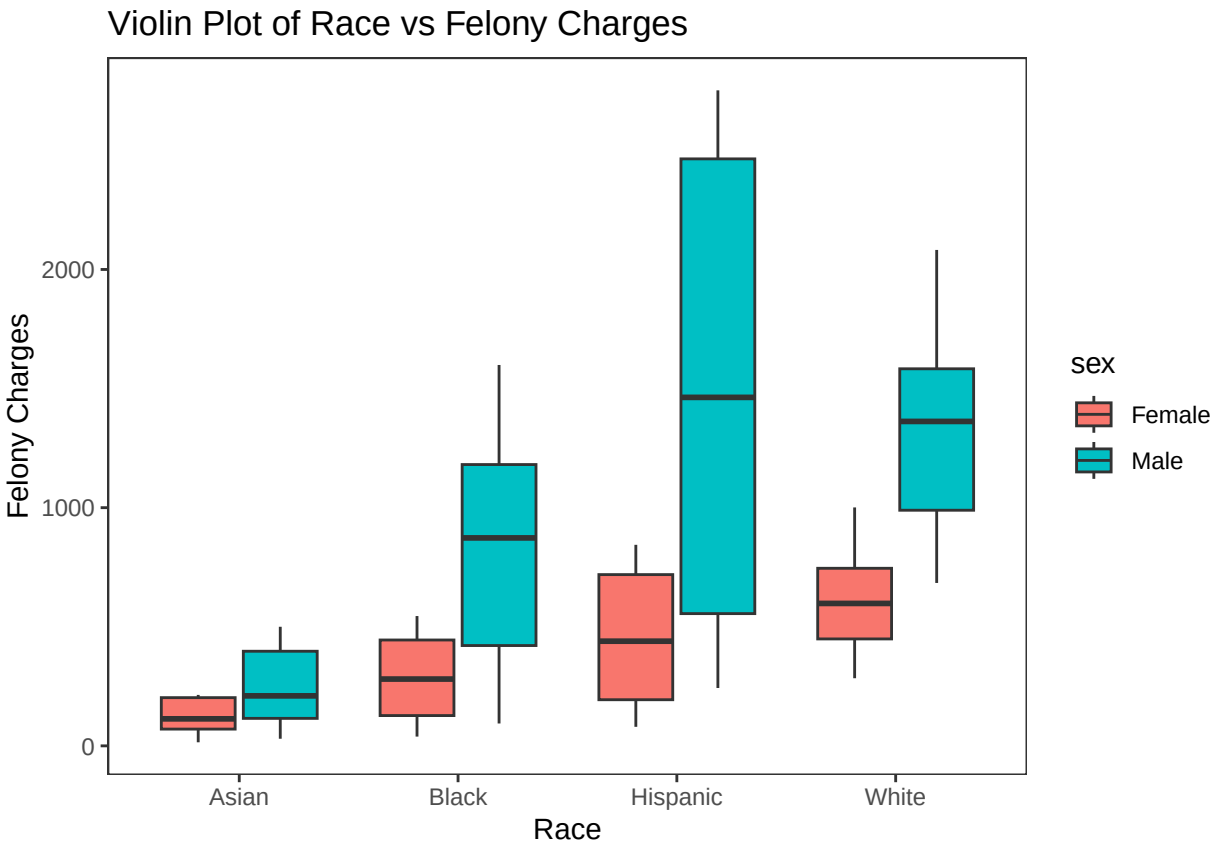

```
# Specifying locations to analyze
dallas_crimes_highway <- filter(dallas_crimes, location == c("Highway, Street, Alley ETC", "Retail Stor
```

```
## Warning in location == c("Highway, Street, Alley ETC", "Retail Store",
## "Apartment Parking Lot", : longer object length is not a multiple of shorter
## object length
```

```
dallas_crimes_highway <- filter(dallas_crimes_highway, sex != "Unknown")

# Plotting the relationship between
ggplot(dallas_crimes_highway, aes(x = age, y = location)) +
  geom_boxplot(fill = "orange") +
  labs(x = "Age", y = "Location") +
  ggtitle("Boxplot of location vs age in Dallas")
```




```
# Plotting the relationship between felony charges and race
race_table <- table(phoenix_crimes$racegrp, phoenix_crimes$felonycharges, phoenix_crimes$sex)
df_race <- as.data.frame(race_table)
colnames(df_race) <- c('race', 'felony', 'sex', 'count')

ggplot(df_race) +
  geom_boxplot(aes(x = race, y = count, fill = sex)) +
  labs(x = "Race", y = "Felony Charges") +
  ggtitle("Violin Plot of Race vs Felony Charges") +
  theme_bw()+ theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank())
```


Violin Plot of Race vs Felony Charges
Felony Charges
2000
1000
0
Asian
Black
Hispanic
White
Race
sex
Female
Male